

PRAKTIKUM SISTEM OPERASI

MODUL 5

Eksplorasi Proses Xinu



Oleh:

Martryatus Sofia

2311104003

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

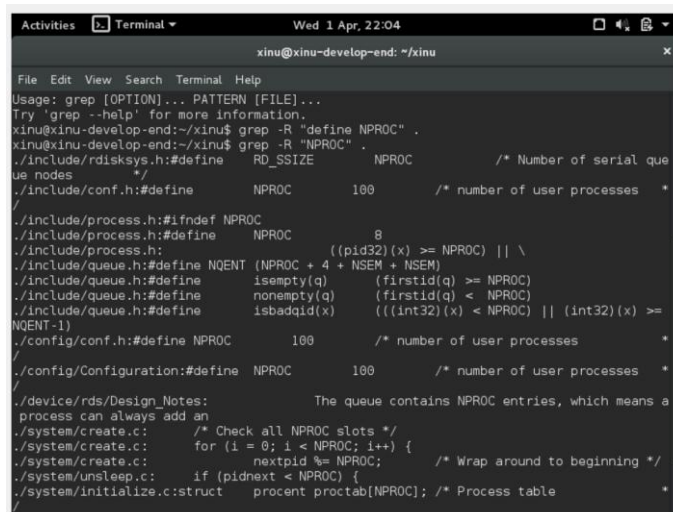
JURNAL

1. Soal 1

[10 Poin] Jawablah pertanyaan berikut ini:

- a. Berapa banyaknya maksimum proses yang ada pada Xinu?

Jawab: 100

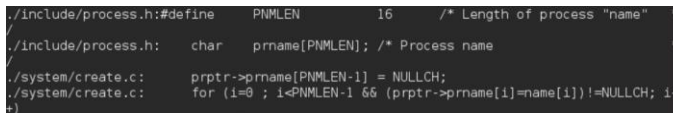


```
Activities Terminal Wed 1 Apr, 22:04
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
Usage: grep [OPTION]... PATTERN [FILE]...
Try 'grep --help' for more information.
xinu@xinu-develop-end:~/xinu$ grep -R "define NPROC" .
./include/rdisksys.h:#define RD_SSIZE NPROC /* Number of serial queue nodes */
./include/conf.h:#define NPROC 100 /* number of user processes */
./include/process.h:#ifndef NPROC 8
./include/process.h:#define NPROC ((pid32)(x) >= NPROC) || \
./include/queue.h:#define NQENT (NPROC + 4 + NSEM + NSEM)
./include/queue.h:#define isempty(q) (firstid(q) >= NPROC)
./include/queue.h:#define nonempty(q) (firstid(q) < NPROC)
./include/queue.h:#define isbadqid(x) (((int32)(x) < NPROC) || (int32)(x) >= NQENT-1)
./config/conf.h:#define NPROC 100 /* number of user processes */
./config/Configuration:#define NPROC 100 /* number of user processes */
./device/rds/Design_Notes: The queue contains NPROC entries, which means a process can always add an
./system/create.c: /* Check all NPROC slots */
./system/create.c: for (i = 0; i < NPROC; i++) {
./system/create.c: nextpid %= NPROC; /* Wrap around to beginning */
./system/unsleep.c: if (pidnext < NPROC) {
./system/initialize.c: struct proctab[NPROC]; /* Process table
```

- b. Berapa maksimal panjang nama suatu proses pada Xinu?

Jawab: 16



```
./include/process.h:#define PNMLEN 16 /* Length of process "name" */
./include/process.h: char pname[PNMLEN]; /* Process name */
./system/create.c: prptr->pname[PNMLEN-1] = NULLCH;
./system/create.c: for (i=0; i<PNMLEN-1 && (prptr->pname[i]!=name[i]); i++)
```

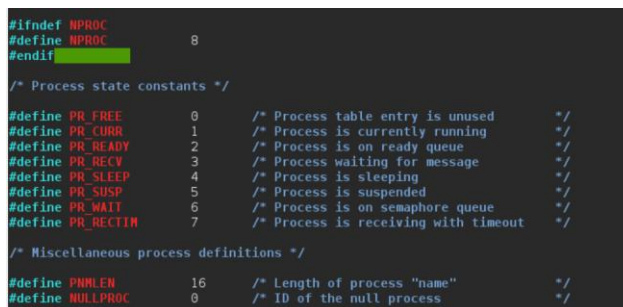
- c. Berapa nilai prioritas awal pada saat proses dibuat?

Jawab: Prioritas awal = parameter priority

`prptr->prprio = priority;`

- d. Ada berapa total state pada Xinu? Sebutkan!

Jawab: 8 state



```
#ifndef NPROC
#define NPROC 8
#endif

/* Process state constants */
#define PR_FREE 0 /* Process table entry is unused */
#define PR_CURR 1 /* Process is currently running */
#define PR_READY 2 /* Process is on ready queue */
#define PR_RECV 3 /* Process waiting for message */
#define PR_SLEEP 4 /* Process is sleeping */
#define PR_SUSP 5 /* Process is suspended */
#define PR_WAIT 6 /* Process is on semaphore queue */
#define PR_RECTIM 7 /* Process is receiving with timeout */

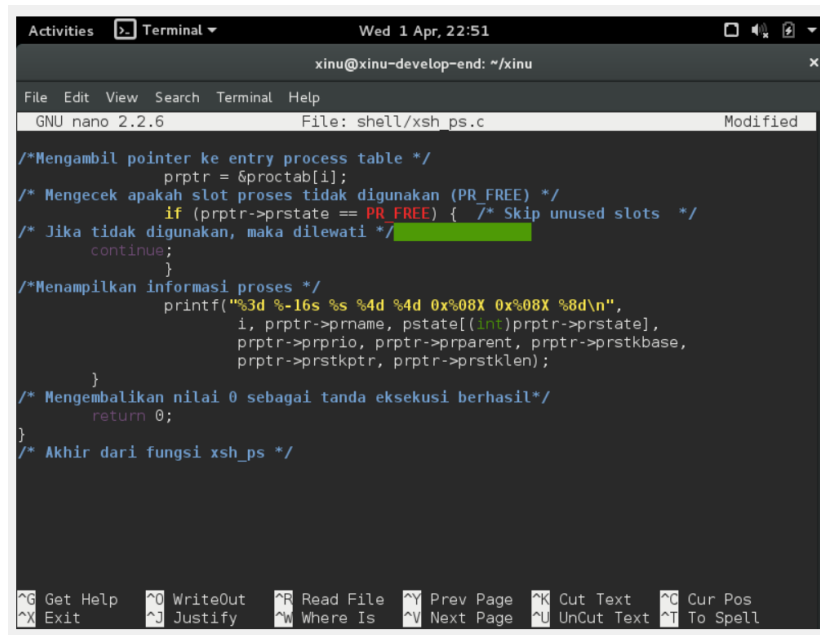
/* Miscellaneous process definitions */
#define PNMLEN 16 /* Length of process "name" */
#define NULLPROC 0 /* ID of the null process
```

2. Soal 2

Perintah ps adalah perintah untuk menampilkan statistik process yang berjalan.

Source code dari ps tersimpan pada file xsh_ps.c. Carilah file tersebut dan beri komentar pada 20 baris terakhir di source code tersebut!

Jawab:



```
Activities Terminal Wed 1 Apr, 22:51
xinu@xinu-develop-end: ~/xinu
File Edit View Search Terminal Help
GNU nano 2.2.6 File: shell/xsh_ps.c Modified

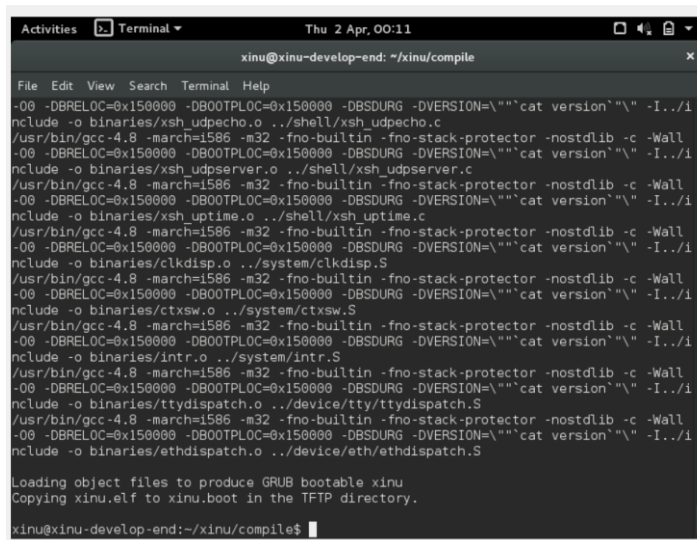
/*Mengambil pointer ke entry process table */
prptr = &proctab[i];
/* Mengecek apakah slot proses tidak digunakan (PR_FREE) */
if (prptr->prstate == PR_FREE) { /* Skip unused slots */
/* Jika tidak digunakan, maka dilewati */
continue;
}
/*Menampilkan informasi proses */
printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
i, prptr->prname, pstate[(int)prptr->prstate],
prptr->prprio, prptr->prparent, prptr->prstkbase,
prptr->prstkptr, prptr->prstklen);
}
/* Mengembalikan nilai 0 sebagai tanda eksekusi berhasil*/
return 0;
}
/* Akhir dari fungsi xsh_ps */
```

3. Soal 3

Ubahlah perintah ps (source code: xsh_ps.c) pada Xinu sehingga menampilkan informasi tambahan berupa kolom yang berisi total message yang ada pada proses seperti gambar di bawah ini: Kolom Msg adalah banyaknya pesan yang ada dalam proses. Kolom Content adalah isi dari pesan tersebut.

Langkah pengerjaan:

- Modifikasi source code pada file xsh_ps.c
- Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya
- Jalankan Backend VM
- Setelah sistem berjalan, jalankan perintah \$ps. Pastikan hasilnya sesuai dengan contoh output pada gambar yang diberikan.
- Screenshot source kode dan output akhir hasil modifikasi



```
Activities Terminal Thu 2 Apr, 00:11
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
-00 -DBRELOC=0x150000 -DBOOTPL0C=0x150000 -DBSDURG -DVERSION="" cat version "\ " -I../i
nclude -o binaries/xsh_udpecho.o ../shell/xsh_udpecho.c
/usr/bin/gcc-4.8 -march=1586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPL0C=0x150000 -DBSDURG -DVERSION="" cat version "\ " -I../i
nclude -o binaries/xsh_udpserver.o ../shell/xsh_udpserver.c
/usr/bin/gcc-4.8 -march=1586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPL0C=0x150000 -DBSDURG -DVERSION="" cat version "\ " -I../i
nclude -o binaries/xsh_uptime.o ../shell/xsh_uptime.c
/usr/bin/gcc-4.8 -march=1586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPL0C=0x150000 -DBSDURG -DVERSION="" cat version "\ " -I../i
nclude -o binaries/clkdsp.o ../system/clkdsp.S
/usr/bin/gcc-4.8 -march=1586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPL0C=0x150000 -DBSDURG -DVERSION="" cat version "\ " -I../i
nclude -o binaries/ctxsw.o ../system/ctxsw.S
/usr/bin/gcc-4.8 -march=1586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPL0C=0x150000 -DBSDURG -DVERSION="" cat version "\ " -I../i
nclude -o binaries/intr.o ../system/intr.S
/usr/bin/gcc-4.8 -march=1586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPL0C=0x150000 -DBSDURG -DVERSION="" cat version "\ " -I../i
nclude -o binaries/ttydispatch.o ../device/tty/ttydispatch.S
/usr/bin/gcc-4.8 -march=1586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPL0C=0x150000 -DBSDURG -DVERSION="" cat version "\ " -I../i
nclude -o binaries/ethdispatch.o ../device/eth/ethdispatch.S
Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.
xinu@xinu-develop-end:~/xinu/compile$
```

Dari jalankan VM Backend masih terkendala tetep grub tapi sebelumnya sudah saya edit semua.

4. SOAL 4

Ubahlah perintah uptime pada Xinu sehingga menampilkan lamanya Xinu sejak booting hanya dalam satuan menit. Langkah pengerjaan:

- Modifikasi source code pada file xsh_uptime.c
- Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya
- Jalankan Backend VM
- Setelah sistem berjalan, jalankan perintah \$uptime. Pastikan hasilnya sesuai dengan contoh output yang diinginkan
- Screenshot source kode dan output akhir hasil modifikasi

```
Activities Terminal Thu 2 Apr, 07:22
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"\"cat version\" \" -I../i
nclude -o binaries/xsh_udpecho.o ../shell/xsh_udpecho.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"\"cat version\" \" -I../i
nclude -o binaries/xsh_udpsvr.o ../shell/xsh_udpsvr.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"\"cat version\" \" -I../i
nclude -o binaries/xsh_uptime.o ../shell/xsh_uptime.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"\"cat version\" \" -I../i
nclude -o binaries/clkdsp.o ../system/clkdsp.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"\"cat version\" \" -I../i
nclude -o binaries/ctxsw.o ../system/ctxsw.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"\"cat version\" \" -I../i
nclude -o binaries/intr.o ../system/intr.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"\"cat version\" \" -I../i
nclude -o binaries/ttydispatch.o ../device/tty/ttydispatch.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=\"\"cat version\" \" -I../i
nclude -o binaries/ethdispatch.o ../device/eth/ethdispatch.S

Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.

xinu@xinu-develop-end:~/xinu/compile$
```

Dari jalankan VM Backend masih terkendala tetep grub tapi sebelumnya sudah saya edit semua.